Exercise 3 (30%)

Kudła et al. (2023) estimated ordered logit models for inheritance and gift taxation among the EU countries. They "analyzed the impact of a series of factors, such as the country's affluence, political preferences, preferences for equity, aging ratio, fiscal standing of the state, and the country's size, on inheritance tax systems". One of the research hypotheses was:

"Hypothesis 4: High inheritance and gift taxation depend positively on the proportion of elderly people to young people in a society."

Another research hypothesis was:

"Hypothesis 5: In highly populated [Big] countries the taxation of inheritance and gifts is higher."

The dependent variable quantified different types of tax designs. The "levels were coded as cardinal numbers from 0 (no tax) to 3 (progressive tax with exemptions) and used as the dependent variable".

"The data about the citizens' wealth were taken from the Credit Suisse Global Wealth Databook. The political variables were obtained from the V-Dem Dataset, version 11.1 (Varieties of Democracy) [Lührmann et al., 2020; Pemstein et al., 2020]. The other variables, including demographic, economic, and financial indicators, were gathered from Eurostat."

"The Independence of youth is a percentage of the population aged between 18 years and 34 years and living with parents. The variable measures the need for the attention of parents (which should be lower for a higher share of children living together with parents) and the possible attractiveness of the financial support delivered by parents to their children. The cost of attention should be lower when young people live together with their parents and conversely, higher independence requires higher gift or inheritance promises. A high share in the population of persons age 75 or higher (75+ share) is a proxy for the potential number of wealth transfers transferred to young individuals. The Aging ratio is the proportion of people aged greater than 64 years to those aged less than 21 years. The greater value of this ratio indicates a population aging and a decrease in the bargaining power of parents (the attention is less available and with higher cost because the number of young people is insufficient to provide attention to older people). The perspective of inheritance should also be affected by life expectancy, which is approximated by the expected number of years of future life at birth (Life expectancy). Finally, we add the sign of net migration, Migration (sign). If immigration was higher than emigration, then this variable took the value 1, and otherwise it took 0. We use the sign of migration instead of immigration and emigration difference to obtain convergence in the estimation of parameters. The net inflow of people should alleviate the problem of the limited attention provided by the younger population to their parents as it increases the number of people who can provide care to older people. It simultaneously decreases the meaning of inheritances and gifts, which alleviates the need for complicated and burdensome inheritance taxation. The second group of variables refers to the wealth of a country and includes gross domestic product per capita (GDP per capita), gross domestic product per capita in purchasing power standard (GDP per capita in PPS), and wealth per adult (Wealth per adult). The first two variables approximate the affluence of the country and the third refers to the accumulated wealth of a representative citizen. They can be interrelated, and so we apply them in separate econometric models. The third group of variables consists of some macroeconomic variables, like Inflation measuring the consumer price index."

"The policy orientation of the state is captured by three variables (Varieties of Democracy data [V-Dem Dataset, version 11.1]):

- orientation of the ruling party (Economic left-right scale), defined on a scale from 0 to 5. The scale starts at far left (0), through left (1), center left (2), center (3), center right (4), and right (5), to far right (6), where left means a higher preference for the active role of government in the economy; and
- (3) the illiberalism index (Illiberalism) taken from V-Dem, which measures the "extent to which the party shows a lack of commitment to democratic norms prior to elections." It represents the measure of populism in a society. The variable takes values in the range 0–1."

"Finally, we add a dummy variable (Big countries) to distinguish between the more populated countries and the other ones. If the country in question is France, Germany, Italy, Poland, Spain, or the United Kingdom, this variable takes on a value of 1, and it takes 0 otherwise. The larger countries easier enforce the payment of inheritance taxes and can benefit more from large inheritances (high accumulated wealth)."

Use $\alpha = 5\%$ significance level.

a) (5%) Was the ordered logit model a good choice for the dependent variable? Explain
b) (5%) Should we use a simple logit instead of the ordered logit for types of tax designs? Explain.
c) (5%) Would a Poisson regression model be an appropriate choice of types of tax designs? Explain.
d) (5%) Interpret the estimate for *Big countries*.
e) (5%) Interpret the estimate for *Aging ratio*.
f) (5%) List diagnostic tests that can be used for ordered choice models.

a) (5%) Was the ordered logit model a good choice for the dependent variable? Explain
b) (5%) List all significant variables in model (1).
c) (5%) Which model should be selected as superior out of those presented? Explain.
d) (5%) How to verify Hypothesis 4? Explain.
e) (5%) How to verify Hypothesis 5? Explain.
f) (5%) List diagnostic tests that can be used for ordered choice models.

Dependent: Tax design type	**(1)**	**(2)**	**(3)**	**(4)**
Independence of youth	0.05	0.06	0.05	0.04
	(0.05)	(0.05)	(0.05)	(0.05)
75+ share	281.55***	284.99***	279.08***	248.07***
	(83.62)	(79.81)	(79.52)	(76.30)
Aging ratio	−15.79***	−17.39***	−15.82***	−16.88***
	(4.96)	(4.90)	(4.70)	(4.77)
Life expectancy	−0.61*	−0.86**	−0.56*	−0.04
	(0.34)	(0.36)	(0.31)	(0.22)
Migration (sign)	−0.37	−0.36	-0.37	−0.28
	(0.86)	(0.87)	(0.87)	0.59
GDP per capita	13.52			
	(59.00)			
GDP per capita PPS		15,485.99		
		(13,033.01)		
Wealth per adult			-0.002	
			(0.088)	
Inflation	−0.15	−0.15	-0.15	
	(0.14)	(0.14)	(0.14)	
Economic left-right scale	0.41	0.40	0.41	0.34
	(0.28)	(0.28)	(0.28)	(0.27)
Illiberalism	−1.80	−1.39	−1.82	−1.17
	(2.19)	(2.22)	(2.19)	(1.97)
Big countries#	15.82***	17.21***	16.23***	
	(3.93)	(4.08)	(3.92)	
Countries	27	27	27	27
Joint insignificance (chi-square statistics)	36.74**	37.43**	36.81**	21.63***
Std. dev. of the individual effect	13.29	13.73	13.52	13.62
Individual effects insignificance	380.06***	407.51***	400.03***	463.74***
AIC	403.6	417.5	418.7	407.8
BIC	460.2	510.5	511.7	460.4

#Big countries include France, Germany, Italy, Poland, Spain and the United Kingdom.

*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

GDP, gross domestic product; PPS, purchasing power standard, AIC, Akaike Information Criterion, BIC, Bayesian Information Criterion.